SLIIT UNI
THE KNOWLEDGE UNIVERSITY

Faculty of Computing

Year 2 Semester 1 (2026)

IT2120 - Probability and Statistics — Lab Sheet 03

The data set is about accommodations of set of students who are studying at University College.

X1 = Age of Student
X2 = Gender
- 1 = Male
- 2 = Female

X3 = Accommodation
- 1 = Stays at Home
- 2 = Boarded Students
- 3 = Lodging

Before starting the lab sheet, you need to create a folder in your desktop and save all your working inside the folder. Set the working directory to that folder using the following command:

setwd("paste the path of the folder")

Eg:- setwd("D:2025 - Sem 2IT2120Lab SessionsLab 03")

1. Identify the variables and import the given data set into R.

```
##Part 01
#Importing Data set
data<-read.csv("DATA 3.csv",header=TRUE)
#View the file in a separate window
fix(data)
#Renaming column headings
names(data)<-c("Age","Gender","Accommodation")
#Converting Gender & Accommodation into factors/categorical variables
data$Gender <- factor(data$Gender, c(1,2), c("Male","Female"))
data$Accommodation <- factor(data$Accommodation, c(1,2,3), c("Home","Boarded","Lodging"))
#Attach the file into R. So, you can call the variables by their names
attach(data)
```

2. Analyze the data in a single variable at a time (univariate analysis). [You should know the suitable technique which needs to apply for each variable to analyze them separately.]

```
##Part 02
#One-way Frequency Table
gender.freq <- table(Gender)
gender.freq

acc.freq <- table(Accommodation)
acc.freq

#Bar chart
barplot(gender.freq, main = "Bar Chart for Gender", ylab = "Frequency",xlab="Gender")
abline(h=0)

barplot(acc.freq, main = "Bar Chart for Accommodation", ylab = "Frequency",xlab="Accommodation")
abline(h=0)

#Pie Chart
pie(gender.freq, main = "Pie Chart for Gender")

pie(acc.freq, main = "Pie Chart for Accommodation")

#Histogram
hist(Age,main="Histogram for Age")

#Box plot
boxplot(Age,main="Box Plot for Age",horizontal=TRUE,outline=TRUE)
```

3. Analyze gender and accommodation together (bivariate analysis). [You should know the suitable technique which needs to apply to analyze two categorical variables simultaneously.]

```
##Part 03
#Two way Frequency Table
gender_acc.freq <- table(Gender,Accommodation)
gender_acc.freq

# Component / Stacked bar chart
barplot(gender_acc.freq, beside=FALSE, main = "Gender & Accommodation",legend = rownames(gender_acc.freq),
        xlab="Accommodation",ylab="Frequency")
abline(h=0)

# Multiple / Clustered bar chart
barplot(gender_acc.freq, beside = TRUE, main = "Gender & Accommodation",legend = rownames(gender_acc.freq),
        xlab="Accommodation",ylab="Frequency")
abline(h=0)
```

4. Analyze age with gender/accommodation. [You should know the suitable statistical technique which needs to apply for this.]

```
# Part 04
# Side-by-Side boxplots
boxplot(Age~Gender, main = "Boxplots for Age by Gender", xlab = "Gender", ylab = "Age")
boxplot(Age~Accommodation, outpch = 8, main = "Boxplots for Age by Accommodation", xlab = "Accommodation",
        ylab = "Age",outline=TRUE)
```

5. Find the mean age for all the combinations of gender and accommodation.

   Hint: Use following format.

		Accommodation		
		Home	Boarded	Lodging
Gender	Male	Mean Age	Mean Age	Mean Age
	Female	Mean Age	Mean Age	Mean Age

```
# Part 05
# 2 categorical with a numerical
xtabs(Age~Gender+Accommodation)/gender_acc.freq

detach(data)
```

Exercise

Instructions: Create a folder in your desktop with your registration number (Eg: "IT......"). You need to save the R script file and take screenshots of the command prompt with answers and save it in a word document inside the folder. Save both R script file and word document with your registration number (Eg: "IT....."). After you finish the exercise, zip the folder and upload the zip file to the submission link.

1. Import the dataset ('Exercise.csv') into R and store it in a data frame called "student_data".
2. Produce the summary statistics and histogram for the variable "X1" (Age).
3. Create a bar chart and frequency table for "X2" (Gender).
4. How does the age (X1) change according to the accommodation (X3)? Analyze it using a suitable graph and interpret the results. (Note that accommodation has three levels which are type 1, type 2 and type 3)